A Machine Learning Based Framework for Credit Card Fraud Detection: Comparative Study of Resampling and Classification Techniques

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Ashwanth Sathish
MSc. Computing (Artificial Intelligence)
Dublin City University
Dublin, Ireland
ashwanth.sathish2@mail.dcu.ie

Chandana Daggupati
MSc. Computing (Data Analytics)
Dublin City University
Dublin, Ireland
chandana.daggupati2@mail.dcu.ie

Sandya Jalesh Kumar
MSc. Computing (Artificial Intelligence)
Dublin City University
Dublin, Ireland
sandya.jaleshkumar2@mail.dcu.ie

Venga Vamsi Krishna Maanam
MSc. Computing (Artificial Intelligence)
Dublin City University
Dublin, Ireland
vengavamsikrishna.maanam2@mail.dcu.ie

***Abstract*—Credit card fraud, including card-not-present (CNP) transactions, identity theft, and transaction-based manipulation, continues to escalate as a global threat. In 2024 alone, over 269 million credit card records were exposed, contributing to an estimated $30 billion in losses worldwide. Despite widespread adoption of fraud detection systems, these tools often struggle with high false positive rates, operational inefficiencies, and lack of interpretability, which hinders their effectiveness in real-time financial environments. This study introduces a fraud detection framework that minimizes false positives and false negatives in credit card transactions. By leveraging feature engineering and feature selection and systematically evaluating multiple resampling strategies, the proposed model enhances fraud detection performance while maintaining scalability and practical deployment readiness.**

***Index Terms*—Credit card fraud detection, Imbalanced classification, Oversampling techniques, Feature engineering, CRISP-DM**

I. Introduction

The widespread adoption of credit cards and the exponential rise in digital transactions have brought unprecedented convenience to consumers and businesses. However, this has also fueled a rapid surge in credit card fraud incidents globally. Global credit card fraud losses are projected to surpass $43 billion by 2026, with the United States alone accounting for nearly 46% of these losses. As illustrated in Fig. 1, both the total fraud value and the fraud cost per $100 spent have steadily increased over the last decade, highlighting the financial magnitude and persistence of the threat. These trends underscore the urgent need for intelligent and adaptive fraud detection systems capable of operating effectively in real-time and high-volume transactional environments.

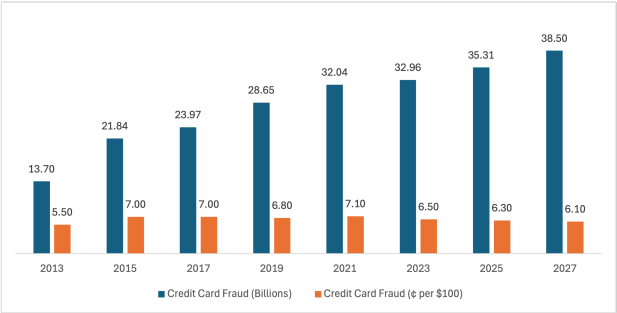

Fig. 1. Global Credit Card Fraud Trends (2013–2027): Total Losses (Billions USD) and Cost per $100 (in Cents)

Financial institutions are under increasing pressure to detect fraudulent transactions accurately while minimizing false positives that disrupt legitimate customer experiences. This challenge is compounded by the extreme class imbalance typical of fraud datasets, where fraudulent transactions often comprise less than 1% of the total records. This imbalance hampers traditional classifiers, which tend to favor the majority class, leading to poor recall on the minority (fraudulent) class.

Deep learning (DL) models have gained increasing attention in fraud detection due to their capacity to model complex and nonlinear transaction patterns. One proposed approach leveraged convolutional and recurrent neural networks with extensive hyperparameter optimization, achieving high accuracy and F1-scores [1]. Despite strong predictive performance, this method did not incorporate mechanisms to handle class imbal-

ance and computational efficiency, both of which are essential for real-world deployment in sensitive financial environments.

Addressing class imbalance has been shown to significantly improve the performance of fraud detection models. One study demonstrated that combining SMOTE (Synthetic Minority Oversampling Technique) with AdaBoost can lead to substantial gains in recall and F1-score across multiple machine learning classifiers [2]. While these oversampling techniques are effective, they are not without limitations such as the risk of introducing synthetic noise or disrupting temporal transaction patterns. These trade-offs highlight the importance of a comparative evaluation of resampling strategies to identify optimal configurations for different model types and datasets.

This paper is motivated by the need to build a fraud detection system that's not only accurate, but also transparent and practical for real-world use. To tackle challenges like class imbalance and model interpretability, we explore several machine learning models logistic regression, random forest, and XGBoost alongside sampling techniques like SMOTE, RUS (Random Under Sampling), and ADASYN (Adaptive Synthetic Sampling). Section II reviews related research and highlights key gaps. Section III explains our approach using the CRISP-DM framework. In Section IV, we evaluate model performance, compare with past studies in Section V, and wrap up with conclusions and future work in Section VI.

II. RELATED WORK

Al-Hashedi and Magalingam [3] provided a wide ranging review of data mining approaches applied to financial fraud between 2009 and 2019. The paper systematically catalogs over a decade of techniques, such as decision trees, logistic regression, Bayesian models, and support vector machines. While the comprehensiveness is valuable, the review is largely descriptive and does not incorporate critical comparisons or evolving DL techniques. As a foundational work, its strength lies in mapping the historical progression of fraud detection models, but it lacks insight into modern, real-time, or scalable solutions.

Bathla and Kaur [4] tackled one of the most pressing issues in fraud detection class imbalance by combining SMOTE with AdaBoost across a variety of machine learning classifiers. Their experiments clearly demonstrate that oversampling the minority class followed by boosting significantly improves recall and F1-score. Particularly notable is their analysis of how these methods impact different classifiers differently. However, the paper acknowledges that SMOTE can introduce synthetic noise and fail to preserve the temporal nature of real transactions. Also, the authors do not explore cost-sensitive learning, which is often crucial when false positives carry business consequences. Still, this study provides one of the most practical and replicable strategies for handling imbalance and should be considered a methodological cornerstone.

Amiri et al. [5] investigated autonomous learning techniques for complex pattern recognition in interconnected systems. While not strictly focused on financial fraud, the architectures they propose particularly those involving reinforcement learning and unsupervised adaptation have meaningful parallels. The paper is rich in theoretical exploration but lacks empirical validation using financial datasets. As such, while the methodologies are cutting-edge, their relevance to real-world credit card fraud detection is speculative and would benefit from applied follow-up studies.

Ileberi et al. [6] presented a credit card fraud detection system that utilizes genetic algorithms for feature selection, integrated with machine learning classifiers like SVM (Support Vector Machine) and decision trees. This combination effectively improved accuracy and reduced false positives. A key contribution is their focus on dimensionality reduction and model interpretability through selected features. However, GA introduces computational cost and requires precise parameter tuning, which can be impractical in high speed or streaming transaction environments. The study performs well on static datasets but leaves questions unanswered regarding model drift and adaptability over time.

Hafez et al. [7] offered one of the most comprehensive and critical systematic reviews in recent literature, analyzing AI-enhanced fraud detection strategies across both ML and DL domains. Their strength lies in not only summarizing trends like the rise of hybrid and ensemble methods but also critiquing the over reliance on outdated datasets like the UCI repository. They call out the lack of standardization in evaluation metrics and highlight how methodological opacity prevents real-world reproducibility. Importantly, this paper introduces a research agenda emphasizing interpretability, dynamic model retraining, and industry collaboration. It acts as a pivotal reference point for both new researchers and system designers seeking to bridge academic insight and operational deployment.

Bansal et al. [8] conducted a comparative analysis of common ML classifiers KNN (K-Nearest Neighbors), SVM, decision trees, genetic algorithms, and LSTM on a benchmark dataset. Their key finding is that LSTM provides superior results due to its temporal learning capability, particularly in sequences of transactions. However, the study lacks evaluation of computational cost, model transparency, and adaptability across datasets, all of which are critical in deployment scenarios. Their analysis is concise but would benefit from more nuanced benchmarking or business-aligned metrics like cost per alert.

Mniai et al. [9] proposed a multi-stage credit card fraud detection framework incorporating undersampling, a hybrid feature selection pipeline, and a Support Vector Data Description (SVDD) model optimized using a custom particle swarm optimization algorithm. This model is particularly adept at maintaining performance even under highly imbalanced conditions. Unlike many papers, they also emphasize scalability and show comparative robustness across different datasets. The downside is the model's complexity and the sensitivity to hyperparameters, which may pose barriers for real-time integration. Nonetheless, this is one of the few technically mature solutions with practical foresight and offers a viable

template for production-grade fraud detection pipelines.

Mienye and Jere [10] authored a focused review on deep learning models used in credit card fraud detection. They discuss the capabilities and trade-offs of CNNs (Convolutional Neural Networks), RNNs (Recurrent Neural Networks), and LSTMs (Long Short-Term Memory). Particularly commendable is their attention to overfitting, model explainability, and computational overhead all critical in fraud analytics. Their inclusion of newer metrics like AUC (Area Under the Curve) and MCC (Matthews Correlation Coefficient) over raw accuracy shows a deep understanding of evaluation challenges in imbalanced classification. However, the review stops short of offering implementation strategies or pipelines for integration. Nevertheless, it is a high-value reference for those evaluating DL architectures for fraud use-cases.

Zioviris et al. [11] introduced a multistage deep learning architecture that improved fraud detection by stacking neural layers to refine prediction quality. The multistage approach helped in feature enhancement and improved the model's ability to generalize, especially in the presence of sparse fraud events. However, the black-box nature of the model and the lack of interpretability mechanisms limit its trustworthiness. Furthermore, deployment considerations such as memory load or latency are not explored, which are essential in real-time applications.

Alarfaj et al. [12] compared multiple ML and DL models, combining them in a layered architecture to detect fraudulent transactions. Their results showed that hybrid models particularly ensembles of tree-based and neural models delivered robust performance across datasets. However, their evaluation focused primarily on traditional metrics without considering real-world deployment metrics like alert fatigue, customer impact, or investigation cost. The paper is comprehensive in scope but stops short of offering deployment recommendations or handling dynamic fraud tactics.

Habibpour et al. [13] made a novel contribution by introducing uncertainty-aware models in fraud detection using Monte Carlo dropout and deep ensembles. This allows the system to output not just predictions but also confidence levels, a significant step toward trustworthy AI in financial systems. The model achieved state-of-the-art results while reducing overconfidence in misclassified instances. However, the paper also notes the computational burden introduced by ensemble predictions, making this approach best suited to high-value or batch-evaluated transactions rather than high-frequency streaming use-cases. Still, their emphasis on predictive certainty is a major advancement in fraud analytics.

Ryman-Tubb et al. [14] conducted one of the rare industry-aligned studies, benchmarking academic models against operational business metrics such as alert rate, manual review burden, and fraud savings. They argue persuasively that most research overlooks cost-sensitive modeling, human interpretability, and regulatory compliance all of which are essential for practical deployment. Their analysis of real-world constraints, including organizational policies and data sharing regulations, makes this paper essential reading for practitioners and academic researchers alike. Importantly, they call for industry-standard metrics and datasets to align scholarly innovation with real-world utility. Roseline et al. [15] proposed an LSTM-RNN based autonomous detection system with an attention mechanism to prioritize features dynamically. Their model achieves high precision and outperforms several baselines.

III. Data Mining Methodology

This study adopts the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework to guide the end-to-end development of a credit card fraud detection system. The framework's iterative and structured nature made it well-suited to managing the complexity of a large, imbalanced dataset while ensuring model transparency and practical applicability.

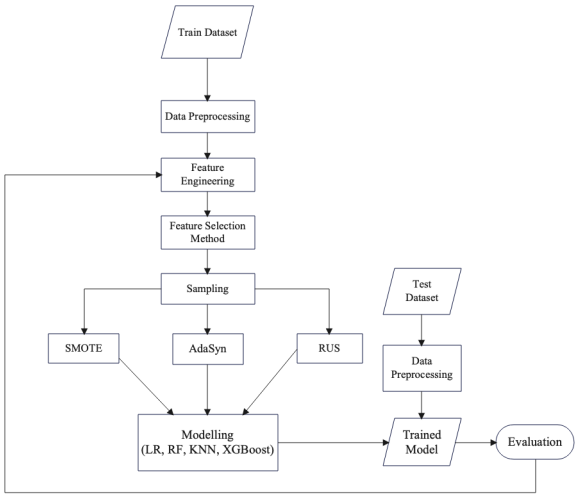

Fig. 2. Framework for Model Development Using Sampling Techniques.

A. *Business Understanding*

Credit card fraud poses a significant threat to financial institutions and consumers, not only resulting in substantial financial losses but also damaging customer trust. The goal of this study is to develop a machine learning pipeline that can detect fraudulent credit card transactions with high accuracy, low false negatives, and clear interpretability. These objectives are captured through the use of the F1-score and AUC as primary evaluation metrics. The F1-score balances precision and recall to reflect both types of misclassification risks, while the AUC captures the model's overall ability to discriminate between fraudulent and non-fraudulent transactions. These objectives are essential in real-world scenarios where wrongly classifying fraud (false negatives) can result in financial damage, and wrongly flagging legitimate transactions (false positives) can erode user experience.

B. *Data Understanding*

The dataset used in this study was sourced from a publicly available Kaggle dataset titled Credit Card Fraud Detection [16], containing approximately 1.29 million transaction

records collected over a two-year period (Jan 2019 – Dec 2020). This synthetic dataset was generated using the Sparkov Data Generation tool and simulates the behavior of 1,000 credit card users across 800 merchants. It includes both legitimate and fraudulent transactions, enabling supervised binary classification.

Initial exploration confirmed the presence of significant class imbalance, with only 0.58% of records labeled as fraudulent. The dataset included transaction details such as timestamps, merchant metadata, geographic information, and user demographics. Exploratory Data Analysis (EDA) identified key behavioral differences between fraudulent and non-fraudulent activity, including anomalies in transaction frequency, amount, and timing. These insights informed the design of several feature engineering and resampling strategies applied in subsequent steps.

C. *Data Preparation*

As part of the data cleaning process, all rows containing missing values were identified and dropped, as the volume was minimal and would not impact model quality. Duplicate records were removed to prevent data leakage. Categorical features were processed based on their cardinality: high-cardinality fields (e.g., merchant, profession, region) were frequency encoded for scalability, while low-cardinality variables (e.g., transaction type, gender) were one-hot encoded to preserve clarity. Boolean fields were converted into binary integers, and the target label indicating fraud status was encoded as 0 (non-fraud) or 1 (fraud) for binary classification compatibility. Significant preprocessing and feature engineering were conducted to improve model performance:

- **Datetime Features:** Transaction times were decomposed into transaction hour, day of month, day of week, weekend indicator, and nighttime indicator to capture temporal behavior. Further, analysis of fraudulent transactions by time of day revealed a significant concentration between 10 PM and 3 AM. As shown in Fig. 3, the majority of fraud occurs during these late-night hours, supporting the inclusion of hour-based features. This aligns with insights from Stripe and Merchant Cost Consulting [17], which note that fraud often occurs outside regular business hours, when monitoring is limited and user response is delayed. These findings justified the inclusion of hour-based features and helped improve temporal sensitivity in our models.
- **Behavioral Patterns:** Features such as daily, weekly, and monthly transaction frequencies were created to reflect card usage patterns. As shown in the boxplot (Fig. 4), fraudulent transactions tend to occur on accounts with significantly lower monthly activity, suggesting fraudsters may target dormant or underused cards. This observation is supported by previous researches [18] that found that fraudulent behavior often emerges in accounts with atypical or infrequent usage patterns, emphasizing the importance of transaction frequency features in fraud modeling.

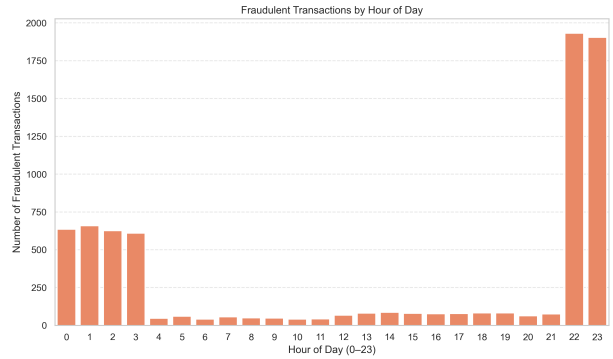

Fig. 3. Distribution of fraudulent transactions across different hours of the day.

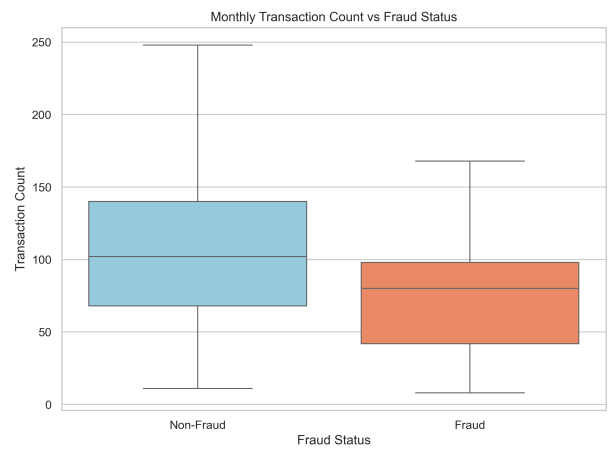

Fig. 4. Monthly distribution of fraudulent versus legitimate transactions.

- **Statistical Features:** A set of transaction level metrics was created to quantify deviation from a cardholder's normal behavior. These features enable the model to detect subtle anomalies that may not be apparent from absolute values alone. The following statistical features were derived:
  - **Average transaction amount per card:**

$$\text{avg_amt} = \frac{\sum_{i=1}^{n} \text{amt}_i}{n}$$

    where amt_i is the amount of the i-th previous transaction and n is the number of past transactions for a given card.
  - **Standard Deviation:**

$$\text{std_amt} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\text{amt}_i - \text{avg_amt})^2}$$

    This measures how much transaction amounts deviate from the cardholder's average.
  - **Z-score:**

$$z = \frac{\text{amt} - \text{avg_amt}}{\text{std_amt}}$$

where amt is the amount of the current transaction. The Z-score quantifies how unusual the transaction is relative to the customer's typical behavior. A higher Z-score indicates a larger deviation from the norm, increasing the likelihood that the transaction is fraudulent. This was found to significantly influence the model (p-value < 0.00001), making it a key indicator of anomalous behavior.

- **Rolling Window Features:** 7 day rolling total amount and transaction count were computed per card using a sliding time window grouped by card number and transaction date. These features helped identify sudden surges in transaction behavior a pattern often observed in fraud bursts. While not essential for all models, their inclusion improved performance in tree based and distance based classifiers by capturing short term behavioral anomalies.
- **Distance Metrics:** The distance between a cardholder's home location and the merchant's location was calculated using geospatial coordinates. While not always significant on its own, this feature was valuable in identifying transactions in unusual locations and worked well alongside other behavioral indicators for detecting fraud.
- **Time Gap Features:** Time elapsed since the previous transaction and its categorized form identified rapid repeat activity typical in fraud bursts. A bar plot (Fig. 5) showed that nearly 70% of frauds happen within 6 hours of the previous transaction, with 44% occurring within the first hour indicating bursty and opportunistic behavior. This aligns with previous research [19] , which has identified time proximity as a critical feature in identifying fraudulent patterns and also suggest that fraudsters often execute multiple transactions in a short period to maximize gains before detection systems can react.

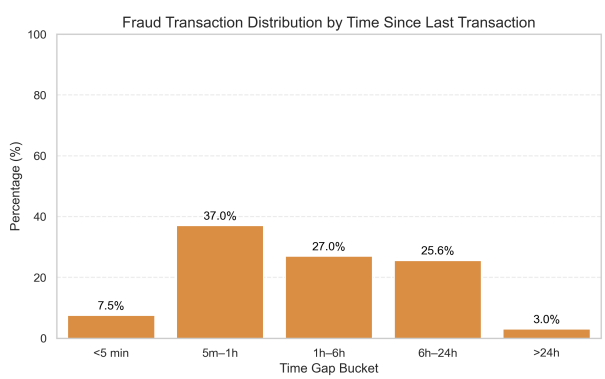

Fig. 5. Distribution of fraudulent transactions based on time since the last transaction on the same card.

- **Fraud Risk Encodings:** Fraud risk scores were computed for both regional and occupational categories by calculating the proportion of fraudulent transactions within each group. These encodings allowed the model to numerically represent contextual risk based on geographic and professional factors, enabling better fraud identification without manual labeling.

D. *Feature Selection*

To reduce dimensionality and retain only the most informative features, we applied a filter-based feature selection method using statistical significance testing. Specifically, we trained a logistic regression model using statsmodels and evaluated the p-values of each feature. Features with p-values greater than 0.05 were considered statistically insignificant and removed. This method provided a computationally efficient way to reduce overfitting and enhance interpretability.

E. *Sampling Strategy*

One of the key challenges in credit card fraud detection is the severe class imbalance, where fraudulent transactions account for less than 1% of all activity. Such imbalance causes standard classifiers to be biased toward the majority class (non-fraud), often leading to poor recall on the minority class. The following oversampling and undersampling techniques were tested:

- **SMOTE (Synthetic Minority Over-sampling Technique):** This method generates synthetic minority class samples by interpolating between existing minority instances. It is widely used due to its effectiveness in creating balanced training sets without discarding any real data.
- **ADASYN (Adaptive Synthetic Sampling):** A variant of SMOTE that focuses on generating synthetic data for harder to learn minority samples, improving model sensitivity in regions with sparse fraud representation.
- **RUS (Random Undersampling):** This method randomly removes samples from the majority class to achieve balance. While simple and computationally efficient, it risks losing important majority class information and may not generalize as well in all models.

All sampling methods were applied only to the training data to avoid data leakage and ensure fair evaluation. After balancing the training set, we trained each classifier on the resampled data and evaluated on the original test set.

F. *Modelling*

Following feature selection based on statistical significance, we trained a range of machine learning classifiers to evaluate their ability to detect fraudulent credit card transactions under varying sampling strategies. The goal was to benchmark models across different learning paradigms: linear, ensemble-based, boost-based, and distance-based, while ensuring interpretability and computational practicality.

The following classifiers were selected:

- **Logistic Regression** was used as a baseline model due to its simplicity, interpretability, and suitability for feature significance testing. It also enabled us to evaluate the impact of key engineered features such as z-score and fraud risk encoding.

- **Random Forest** was chosen for its robustness and ability to model non linear relationships. It consistently demonstrated strong performance across both oversampling and undersampling techniques and benefited from temporal and frequency-based features.
- **XGBoost** a gradient-boosted tree algorithm, was selected for its scalability and accuracy on structured data. It showed particularly strong performance when combined with ADASYN and SMOTE, achieving the highest F1-scores and AUC values in many cases.
- **K-Nearest Neighbors (KNN)** was included to assess performance in a distance based framework. KNN was especially responsive to anomaly based features such as z-score and rolling transaction statistics, although it showed sensitivity to sampling noise.

Each model was evaluated under three sampling strategies: SMOTE, ADASYN, and Random Undersampling. Hyperparameters were kept consistent across resampling methods to isolate the effect of sampling on model performance. All implementations were carried out using scikit learn and XGBoost libraries. Although Support Vector Machines (SVM) and Multilayer Perceptrons (MLP) were initially considered, they were excluded from final experiments due to longer training times and no significant performance advantage in preliminary testing. This allowed the focus to remain on interpretable and computationally efficient models. Model performance was assessed using F1-score and ROC AUC on both a validation set and a separate, independently preprocessed test set to ensure generalization.

IV. EVALUATION

This section presents a comprehensive assessment of model performance, with a focus on the minority class (fraudulent transactions). Due to the dataset's severe class imbalance, we prioritize the F1-score as the most appropriate evaluation metric, as it balances false positives and false negatives. While accuracy is reported for completeness, it is not considered a reliable indicator of success in this context. Precision and recall are also included to provide additional insight into each model's fraud detection capability.

Three resampling strategies, SMOTE, ADASYN, and Random Undersampling (RUS), were evaluated across four classification algorithms: Logistic Regression, Random Forest, XGBoost, and K-Nearest Neighbors (KNN). Performance was assessed on both the training and test sets, using metrics derived from the positive class (fraud only).

A. *Performance Discussion*

Table I and II summarizes the performance of the model in both the training and the test datasets for the fraudulent class (Class 1), grouped by sampling strategy. Among all combinations, XGBoost with SMOTE consistently delivered the best results, achieving an F1-score of 0.928 on the training set and 0.896 on the test set, with high precision and recall maintained across both. This consistency highlights the model's ability to generalize well to unseen data without overfitting. SMOTE enriched the minority class by generating synthetic yet realistic fraud examples, allowing XGBoost to learn diverse patterns without discarding information from the majority class. In contrast, while Random Forest and KNN also benefitted from SMOTE and ADASYN, their test performance was more variable and logistic regression underperformed across all samplers, particularly with RUS, due to low precision. The superior performance of XGBoost was further supported by the robust set of engineered features, such as z-score deviations, rolling transaction metrics, time gap intervals, and fraud risk encodings, which allowed the model to capture nuanced, nonlinear behavior patterns. These features were especially well leveraged by XGBoost's boosting architecture, making the model highly effective in detecting fraudulent transactions.

TABLE I
TRAINING PERFORMANCE OF CLASSIFIERS USING DIFFERENT SAMPLING STRATEGIES (CLASS 1)

Sampling Method	Model	Precision	Recall	F1-Score	Accuracy
SMOTE	Logistic Regression	0.98	0.94	0.96	0.98
	Random Forest	0.94	0.84	0.88	1.00
	XGBoost	0.93	0.93	0.93	1.00
	KNN	0.49	0.86	0.63	0.99
RUS	Logistic Regression	0.08	0.86	0.14	0.94
	Random Forest	0.29	0.97	0.44	0.99
	XGBoost	0.35	0.99	0.52	0.99
	KNN	0.13	0.93	0.22	0.96
ADASYN	Logistic Regression	0.17	0.74	0.28	0.98
	Random Forest	0.95	0.84	0.89	1.00
	XGBoost	0.93	0.92	0.93	1.00
	KNN	0.49	0.86	0.63	0.99

TABLE II
TEST PERFORMANCE OF CLASSIFIERS USING DIFFERENT SAMPLING STRATEGIES (CLASS 1)

Sampling Method	Model	Precision	Recall	F1-Score	Accuracy
SMOTE	Logistic Regression	0.24	0.71	0.36	0.99
	Random Forest	0.96	0.23	0.37	0.99
	XGBoost	0.93	0.87	0.90	0.99
	KNN	0.43	0.79	0.57	0.96
RUS	Logistic Regression	0.06	0.82	0.11	0.95
	Random Forest	0.32	0.81	0.46	0.99
	XGBoost	0.33	0.96	0.49	0.99
	KNN	0.08	0.92	0.14	0.96
ADASYN	Logistic Regression	0.15	0.67	0.24	0.98
	Random Forest	0.94	0.22	0.36	1.00
	XGBoost	0.93	0.87	0.90	1.00
	KNN	0.42	0.78	0.55	1.00

B. *AUC-ROC Curve Comparison*

To evaluate model discrimination beyond a fixed threshold, we plotted the Receiver Operating Characteristic (ROC) curves for all SMOTE-trained models. Fig. 6 displays the AUC results. The ROC curves confirm the superiority of XGBoost, with its curve approaching the top-left corner, indicating strong sensitivity and low false positive rates. These findings rein-

force our F1-based conclusions and demonstrate that XGBoost offers excellent discriminatory power under SMOTE.

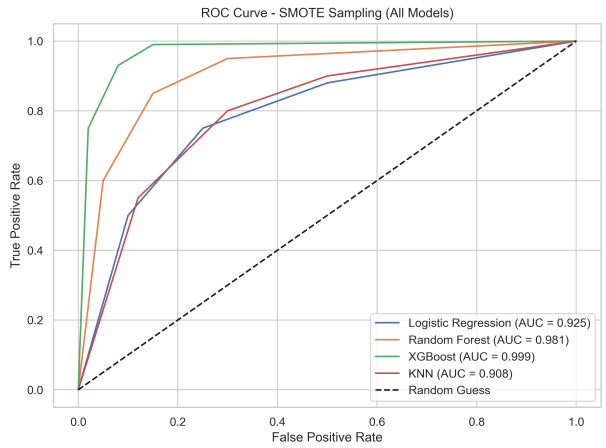

Fig. 6. ROC Curve Comparison Using SMOTE Sampling.

C. *Comparative Benchmark with Existing Studies*

To place our results in context, we compared our best-performing model with prior research that used the same Kaggle dataset.

TABLE III
COMPARISON OF FRAUD DETECTION PERFORMANCE WITH RELATED STUDIES (CLASS 1)

Study	Sampling	Model	Prec.	Rec.	F1	Acc.
Our Study	SMOTE	XGBoost	0.93	0.87	0.90	0.99
Adesina & Howe (2024) [20]	SMOTE+Tomek	RF	0.81	0.69	0.75	0.99
Afriyie et al. (2023) [21]	RUS	RF	0.96	0.17	0.97	0.96

As illustrated in Table III, our model outperforms both prior works in F1-score, which is critical for imbalanced fraud detection tasks. While Adesina & Howe's SMOTE+Tomek configuration achieved strong recall, its precision and overall balance were slightly lower. Afriyie et al.'s model prioritized recall at the expense of precision, resulting in a significantly lower F1-score due to the high false positive rate introduced by random undersampling.

In contrast, our approach benefited not only from optimal model sampling alignment (XGBoost + SMOTE), but also from a comprehensive feature engineering strategy. Features such as transaction frequency, z-score deviation, time gap intervals, and risk based encodings enabled the model to detect nuanced fraud behavior beyond raw transaction values. These domain-specific features improved the separation of legitimate and fraudulent transactions, giving our model an advantage in learning and generalizing complex fraud patterns that earlier studies may have overlooked.

V. CONCLUSION AND FUTURE WORK

This study presents a comprehensive, efficient, and high-performing machine learning pipeline for credit card fraud detection, built using the CRISP-DM methodology. By combining robust preprocessing, domain driven feature engineering, and a comparative evaluation of resampling strategies and classification models, we successfully addressed key challenges in fraud detection namely, class imbalance, false positives and false negatives.

Among the tested models and sampling strategies, XGBoost combined with SMOTE emerged as the most effective configuration, achieving an F1-score of 0.90, precision of 0.93, and recall of 0.87 on the test set. The integration of statistical, temporal, and contextual features significantly enhanced the model's ability to detect nuanced fraud patterns while maintaining transparency through one of the interpretable metric, as p-values. Comparative benchmarking further validated our approach, with our model outperforming existing studies using the same dataset in key fraud detection metrics.

For future work, we aim to incorporate real-time streaming data to enable dynamic fraud detection and reduce latency in production environments. In addition, evaluating cost-sensitive learning strategies may allow us to explicitly minimize the financial impact of misclassifications. Expanding the adaptability of the model through continuous learning would help mitigate concept drift and evolving fraud tactics. We also see value in integrating explainability frameworks, such as SHAP or LIME, directly into analyst dashboards for real time interpretation and audit readiness. Finally, combining our supervised pipeline with unsupervised anomaly detection or graph-based methods could reduce reliance on labeled data and uncover hidden fraud networks.

Overall, the proposed pipeline demonstrates the practical value of combining domain knowledge, statistical rigor, and machine learning for fraud detection systems that are not only accurate, but also reliable and deployable.

The source code used in this study is available at: https://github.com/jsandya7/creditcard-fraud-feature-sampling-study.

ACKNOWLEDGMENT

We express our heartfelt gratitude to Dr. Andrew McCarren, Head of the School of Computing, Faculty of Engineering and Computing, for his exceptional teaching and guidance in Data Analytics and Data Mining. His expertise, support and enthusiasm for the subject greatly contributed to our understanding and inspired the development of this work.

We also extend our appreciation to Dr. Luca Rossetto, Assistant Professor in the Faculty of Engineering and Computing, for his academic support and valuable insights throughout our studies.

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